AI TRANSFORMATION PROPOSAL

Labcorp Central Laboratory Services: Accelerating the Industry 5.0 Transition

Three AI Use Cases | Proposed LC-CLS Governance Framework | Industry 4.0 to 5.0 Roadmap

$5.64B to $8.18B Central Lab Market 2024-30	**86% Oncology** FDA Approvals Supported – LC-CLS 2024	**35-45%** AI Productivity Gain	**6.5% CAGR** Central Lab Market Growth to 2030

1. Executive Summary

Three AI initiatives are proposed for Labcorp Central Laboratory Services (LC-CLS), consistent with Industry 5.0 human-centric principles (Kraaijenbrink, 2022): Protocol Intelligence, synthesising NLP and institutional knowledge; Start-Up Kit Supply Optimisation, shifting from reactive to predictive human-approved supply management; and Automated Gating via OMIQ and Concentriq, all within the LC-CLS AI Framework: industrial AI architecture, ML quality methodology, and EU AI Act requirements.

The three most common AI reservations (accuracy, accountability, and regulatory compliance) are each structurally addressed in Section 3.

Dimension	Detail
Company	Labcorp Central Laboratory Services (LC-CLS), a division of Labcorp
Scale	Global operations; 100+ countries; Phase I–IV clinical trial support
Services	Safety and speciality testing; kit production and logistics; lab trial data and analytics
Regulatory context	GCP-regulated; FDA 21 CFR Part 11; ICH E6(R3); EU AI Act applicability
Competitive position	IQVIA Laboratories (Site Lab Navigator, SAVE tool) (IQVIA, 2025) and Thermo Fisher Scientific (PPD) are deploying AI supply and lab tools; LC-CLS at a strategic inflexion point

Table 1: *LC-CLS Snapshot*

2. Strategic Context: Competitive Pressure and the 5.0 Imperative

The central laboratory services market will reach $8.18 billion by 2030 (CAGR 6.5%; MarketsandMarkets, 2025), whilst Mihic *et al.* (2025) project 35-45% AI productivity gains across clinical development. IQVIA's Site Lab Navigator and SAVE tool ($2.5 million per trial in deferred shipping costs) define the competitive gap (IQVIA, 2025; Clinical Trials Arena, no date).

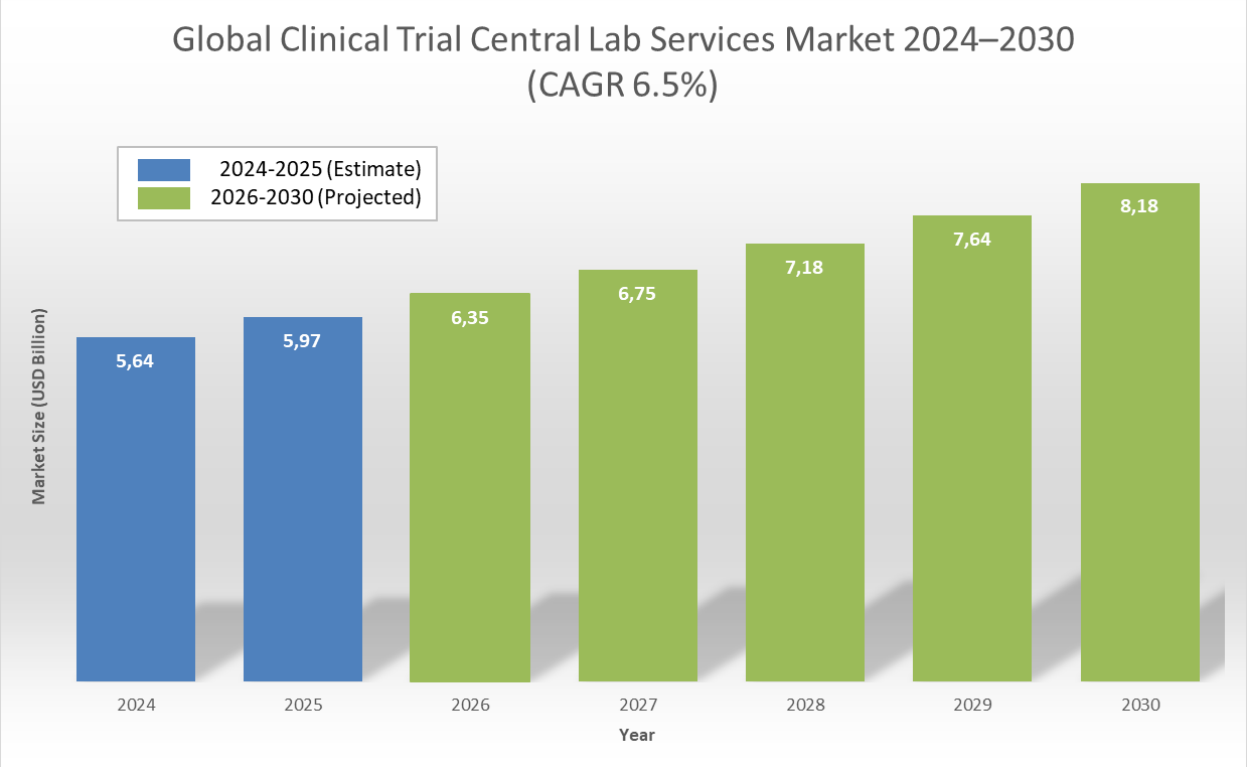


Figure 1: *Central Lab Services Market 2024-2030, CAGR 6.5% (MarketsandMarkets, 2025)*

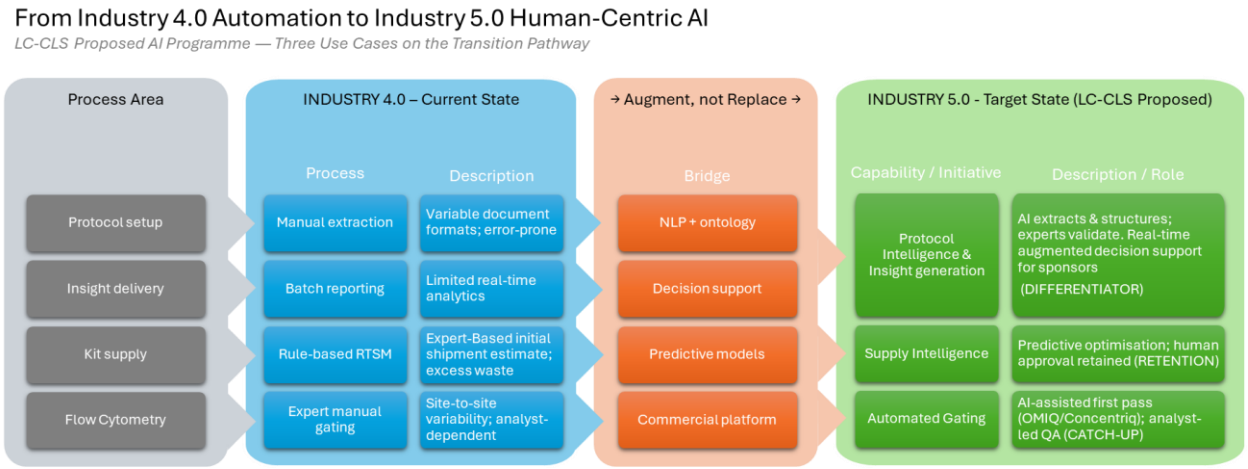


Figure 2: *LC-CLS Transition Pathway: Industry 4.0 to 5.0 (Author; Kraaijenbrink, 2022; Mihic et al., 2025)*

Protocol complexity generates a 76% amendment rate at costs reaching $535,000 per event (Getz et al., 2024). Supply waste is embedded in start-up phase dynamics before enrolment patterns stabilise (Slope Clinical, 2023). Analytical bottlenecks from manual gating create multi-site inconsistency as oncology trial volumes grow (Aghaeepour et al., 2013).

3. Governance Framework Selection

3.1 Candidate Frameworks

Framework	Strengths	Limitations for the GCP context
CRISP-DM	Systematic; widely adopted; effective for exploratory analytics (Schroer *et al.*, 2021)	No production monitoring; no accountability architecture; no change management
CRISP-ML(Q)	Adds quality gates across the ML lifecycle; addresses model performance (Studer *et al.*, 2021)	Agnostic on regulatory accountability; no GCP-specific guidance

NIST AI RMF	Comprehensive risk governance; accountability and bias coverage (NIST, 2023)	No technical implementation detail; not adapted to a regulated laboratory context

Table 2: *Framework Comparison (CRISP-DM / CRISP-ML(Q) / NIST)*

None alone covers the combined operational, regulatory, and accountability demands of a GCP-regulated laboratory.

3.2 The Proposed LC-CLS AI Framework

The LC-CLS AI Framework (Figure 3) synthesises Peres et al. (2020) five technology layers (DT/AT/PT/OT/HT), CRISP-ML(Q) quality gates (Studer et al., 2021), and EU AI Act Article 9 risk management (EU AI Act, 2024) as a structural, not documentary, obligation. The HT layer is non-removable: Leslie (2019) requires accountability traceable to named decision-makers.

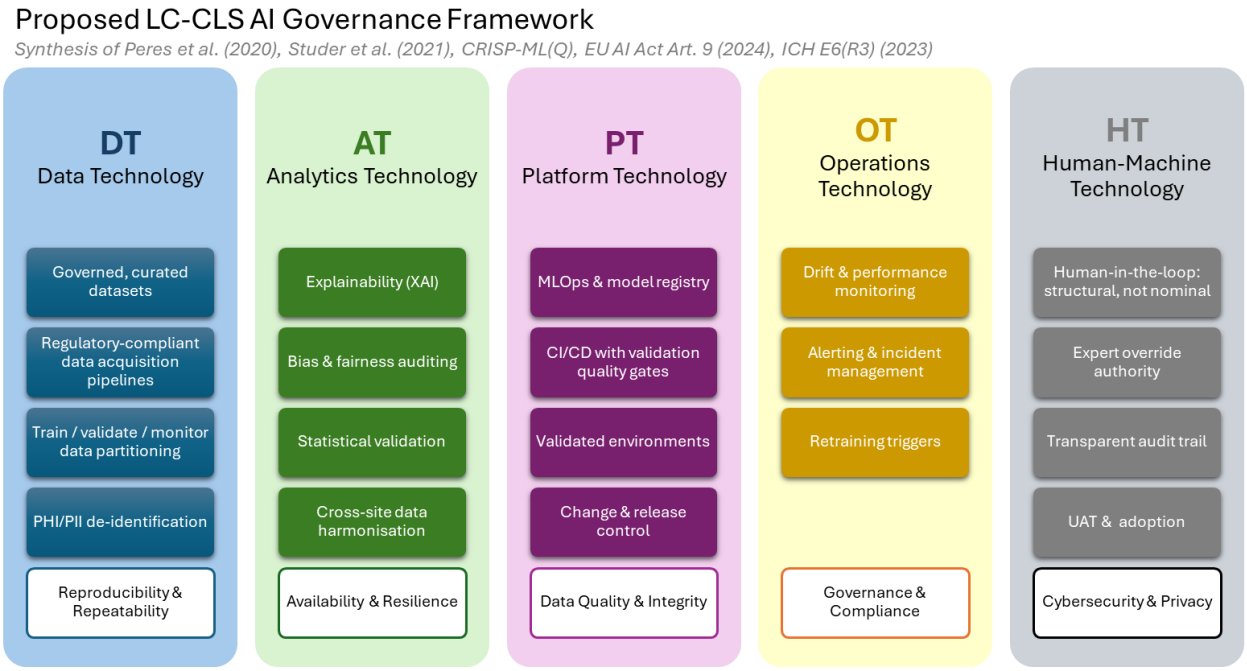


Figure 3: *LC-CLS AI Governance Framework (Author; Peres et al., 2020; Studer et al., 2021; EU AI Act, 2024; ICH, 2025)*

4. Use Case 1: Protocol Intelligence, the Strategic Differentiator

4.1 The Problem

Dimension	Before	After
Workflow	Protocols, RFPs, and quality events held in disconnected systems; no queryable form	AI parses unified corpus against ICH M11; scenario recommendations generated; expert validates all outputs (Leslie, 2019)
Amendment	76% of Phase I-IV trials require a substantial amendment; avg. 260 days per cycle (Getz *et al.*, 2024)	Target: ≥43% reduction in amendable elements flagged pre-SOW
Cost	$141,000-$535,000 per amendment event; errors propagate across the lifecycle (Getz, 2023)	Cost avoidance is proportional to the amendments prevented
Limitation	-	Output quality depends on corpus consistency; poorly structured or incompletely labelled historical records degrade recommendation precision.
Approach	-	RAG retrieves from a live corpus without retraining, preserving institutional knowledge as the study record base grows

Table 3: *Protocol Intelligence - Before/After*

4.2 The Proposed Capability

Protocol Intelligence is a study design advisory system. Ingesting protocols, RFPs, and quality events, it recommends test selection, specimen routing, and kit configuration against ICH M11 (European Medicines Agency, 2025), flagging patterns from past

failures. Experts interrogate, accept or override, in a full audit trail (Leslie, 2019; Kraaijenbrink, 2022). Data stewardship agreements are prerequisite, as historical records may not authorise secondary AI use.

4.3 Differentiation

No competitor holds a comparable advisory corpus for central laboratory study design. Defensibility lies not in the architecture but in the corpus: LC-CLS quality event logs no external vendor holds (Peres et al., 2020). IQVIA and NVIDIA have deployed orchestrator AI agents for trial start-up, performing structured protocol data extraction via specialised sub-agents (Clinical Research News Online, 2025).

5. Use Case 2: Start-Up Kit Supply Optimisation, the Retention Engine

5.1 The Problem

Dimension	Before	After
Workflow	Rule-based RTSM thresholds; reactive replenishment on-site request; no predictive model	Gradient boosting forecast generates a site-specific quantity; Project Manager approves before any shipment.
Waste	~$2.5M excess per trial; competitor (IQVIA SAVE) already operating at this scale (Clinical Trials Arena, no date)	Target: ≥30% reduction in excess kit rate vs pre-deployment baseline
Limitation	-	Gradient boosting performs poorly at newly opened or low-volume sites with sparse enrolment history; minimum data thresholds must be defined before deployment.
Approach	-	Gradient boosting handles non-linear site features; quantile regression delivers

		probabilistic rather than point-estimate forecasts, enabling minimum-excess targeting (Giannopoulos et al., 2025)

Table 4: *Supply Intelligence - Before/After*

5.2 The Proposed Capability

Supply Intelligence applies gradient boosting with a quantile regression objective (Giannopoulos et al., 2025) to forecast site-specific kit quantities with minimum probable excess over predicted recruitment. Input features include site, therapeutic area, study phase, visit schedule, and historical enrolment velocity; the model outputs a recommendation only; project managers retain full approval authority (Kraaijenbrink, 2022).

6. Use Case 3: Automated Gating for Flow Cytometry, the Catch-Up

6.1 The Problem

Dimension	Before	After
Workflow	Specialist analyst manually gates raw FCS files at each site independently; result harmonisation is analyst-dependent	Commercial platform (OMIQ and Concentriq Dp) generates first-pass gate classifications; analyst reviews and adjusts all gates before export.
Variability	Documented inter-analyst inconsistency across distributed global sites (Aghaeepour *et al.*, 2013)	Target: ≥95% concordance vs validated reference panel per monthly QC
Limitation	-	Commercial platforms may not generalise to novel or non-standard assay panels; each new panel requires bespoke fit-for-purpose validation before clinical use.
Approach	Manual classification only	Commercial platform chosen over custom build: GCP requires vendor-qualified

		software with an existing audit trail; bespoke development requires full IQ/OQ/PQ validation from scratch (European Medicines Agency, 2023)

Table 5: *Automated Gating - Before/After*

6.2 The Proposed Solution

OMIQ and Concentriq Dp generates first-pass FCS gate classifications; analysts review and adjust all gates before export, with analyst judgment authoritative throughout (European Medicines Agency, 2023; ICH, 2025). The catch-up label is deliberate: GCP-validated deployment became standard only from 2022; inclusion yields a coherent programme rather than disconnected pilots.

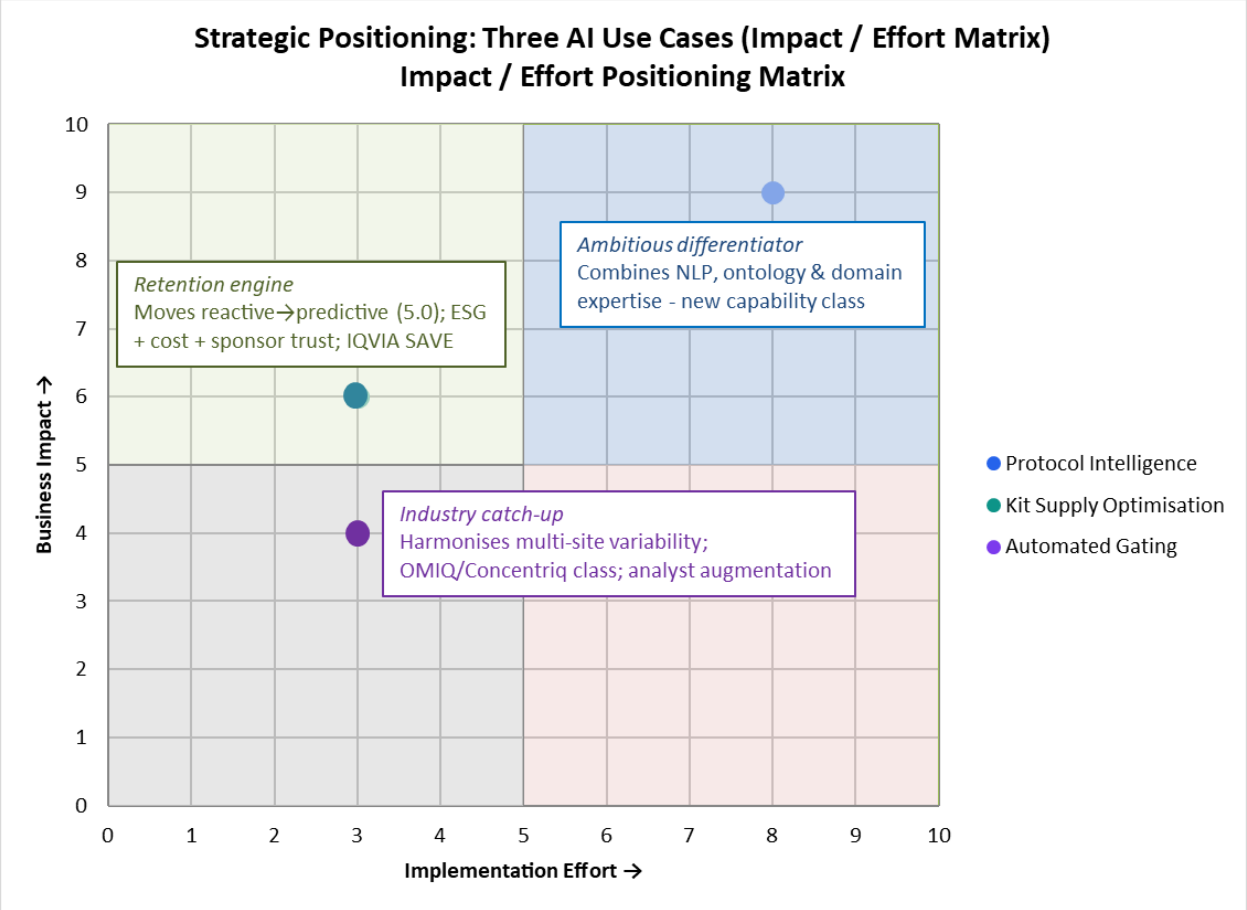


Figure 4: *AI Use Cases Positioning, Impact/Effort Prioritisation (Author; McKinsey & Company, 2025; Getz et al., 2024; Clinical Trials Arena, no date)*

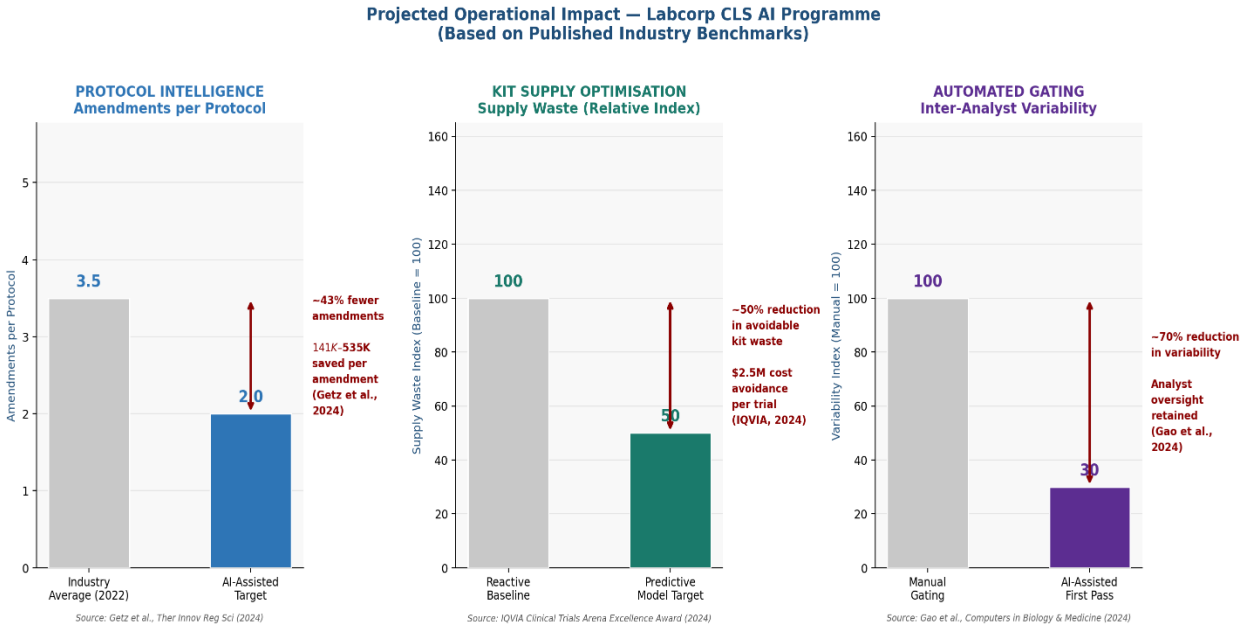


Figure 5: *Projected Operational Impact, LC-CLS AI Programme (Getz et al., 2024; Clinical Trials Arena, no date; Aghaeepour et al., 2013)*

7. Risk Architecture and Change Management

7.1 Change Management and Adoption

Analyst reluctance to log reasoning when acting on algorithmic outputs is a documented adoption barrier requiring active management (Dodig, Čepelak and Dodig, 2025).

Six roles govern deployment: AI Ethics Board (data consent, EU AI Act Art. 9), IT Change Management Board (validated environments), Clinical Operations Lead (process redesign), Project Managers and Senior Analysts (approval and override documentation), and MLOps (drift monitoring).

All users require training on output interpretation, override documentation, and audit obligations before go-live (Leslie, 2019; EU AI Act, 2024).

7.2 Key Risks

Shared IT functions carry finite validated capacity and competing priorities, requiring capacity planning (Dodig, Čepelak and Dodig, 2025). Further risks include data consent constraints, revenue misalignment, validation scope creep, and cybersecurity exposure at AI inference endpoints (EU AI Act, 2024). Where model outputs are acted on without logged reasoning, audit-trail integrity cannot be demonstrated. The Peres et al. (2020) taxonomy originated in manufacturing; its extension to GCP-regulated laboratory services requires empirical validation.

Governance Control	Protocol Intelligence	Kit Supply Optimisation	Automated Gating	Framework Layer	Regulatory Anchor	Impact if Control Fails	Accountability Owner
Human-in-the-loop	Expert validates all AI extractions	PM approves all shipments	Analyst reviews all gates	HT Layer	ICH E6(R3) GCP	Unvalidated outputs enter clinical workflow	Project Manager; Senior Analyst (Gating)
Drift Monitoring	Extraction concordance; quality-event alerts	Forecast vs actual consumption	Concordance vs reference gates	OT Layer	EU AI Act Art. 9	Silent model degradation; errors compound undetected	IT Operations / MLOps Lead
Change Control	Model updates via validated CI/CD pipeline	Threshold changes via structured review	Platform updates per vendor SLA	PT Layer	GCP software validation	GCP bypass; audit gaps; regulatory risk	IT Change Management Board
Audit Trail	Source provenance + extraction logic + decision	Recommendation + override log	Gate suggestions + analyst adjustments	DT/PT Layers	United States (FDA) - 21 CFR Part 11 (2023)	FDA 21 CFR Part 11 non-compliance; data integrity breach	IT Systems Administrator / Data Governance Lead

Ethical Governance	Data ownership and consent for training data	Revenue incentive misalignment	RUO-to-GCP scope boundary	AT Layer	EU AI Act Art. 9	Data misuse; commercial conflict; RUO-to-GCP breach	AI Ethics Board / Legal Counsel

Table 6: *Governance Controls by Use Case*

Use Case	IT KPI	Baseline	Target	Measurement Method
Protocol Intelligence	Protocol extraction concordance rate vs. expert review	Not currently tracked	≥90% concordance on sampled quarterly audit	Blind comparison of AI extraction vs. expert benchmark on historical protocols
Start-Up Kit Supply Optimisation	Forecast mean absolute error vs. actual kit consumption	Manual estimation; no formal MAE baseline	Model MAE ≤15% of actual consumption per study	Monthly model evaluation against dispensing records
Automated Gating	Inter-site concordance vs. validated reference panel	Documented analyst inconsistency (Aghaeepour *et al.*, 2013)	≥95% concordance vs. reference panel	Monthly cross-site QC comparison

Table 7: *IT Performance KPIs*

8. Conclusion: Sustaining Leadership Through Intelligent Augmentation

For LC-CLS, AI adoption is a competitive necessity; the institutional knowledge and regulatory standing to act already exist. The combined value case is quantifiable:

amendment cost avoidance of $141,000–$535,000 per event (Getz et al., 2024) and $2.5 million per trial in excess kit reduction (Clinical Trials Arena, no date). Supply Intelligence closes the IQVIA SAVE gap (Slope Clinical, 2023); Automated Gating provides the multi-site consistency both require (Aghaeepour *et al.*, 2013). Faster amendment cycles accelerate patient access; algorithmic decision support raises workforce implications for change management. Together they form a trustworthy, auditable programme with human authority maintained and EU AI Act obligations embedded structurally (Leslie, 2019; EU AI Act, 2024).

References

Aghaeepour, N. *et al.* (2013) 'Critical assessment of automated flow cytometry data analysis techniques', *Nature Methods*, 10(3), pp. 228–238. Available at: https://doi.org/10.1038/nmeth.2365

Clinical Research News Online (2025) 'IQVIA, NVIDIA build AI agents for clinical research', *Clinical Research News Online*, 11 June. Available at: https://www.clinicalresearchnewsonline.com/news/2025/06/11/iqvia-nvidia-build-ai-agents-for-clinical-research (Accessed: 23-Mar-2026).

Clinical Trials Arena (no date) *Clinical Trials Arena Excellence Awards 2024: IQVIA*. Available at: https://www.clinicaltrialsarena.com/featured-company/2024-iqvia/ (Accessed: 18 February 2026).

Dodig, S., Čepelak, I. and Dodig, M. (2025) 'Are we ready to integrate advanced artificial intelligence models in clinical laboratory?', *Biochemia Medica*, 35(1), 010501. Available at: https://doi.org/10.11613/BM.2025.010501

European Medicines Agency (2023) *Guideline on computerised systems and electronic data in clinical trials*. EMA/INS/GCP/112288/2023. Available at: https://www.ema.europa.eu/en/documents/regulatory-procedural-guideline/guideline-computerised-systems-and-electronic-data-clinical-trials_en.pdf (Accessed: 18 February 2026).

European Medicines Agency (2025) *ICH M11 guideline, clinical study protocol template and technical specifications - Scientific guideline*. Available at: https://www.ema.europa.eu/en/ich-m11-guideline-clinical-study-protocol-template-technical-specifications-scientific-guideline (Accessed: 18 February 2026).

Getz, K. (2023) 'Shining a light on the inefficiencies in amendment implementation', *Applied Clinical Trials*, 6 December. Available at: https://www.appliedclinicaltrialsonline.com/view/shining-a-light-on-the-inefficiencies-in-amendment-implementation (Accessed: 18 February 2026).

Getz, K., Smith, Z., Botto, E., Murphy, E. and Dauchy, A. (2024) 'New benchmarks on protocol amendment practices, trends and their impact on clinical trial performance', *Therapeutic Innovation & Regulatory Science*, 58(4), pp. 539–548. Available at: https://doi.org/10.1007/s43441-024-00622-9

Giannopoulos, P.G., Dasaklis, T.K., Tsantilis, I. and Patsakis, C. (2025) 'Machine learning algorithms in intermittent demand forecasting: a review', *International Journal of Production Research*. Available at: https://doi.org/10.1080/00207543.2025.2578701 (Accessed: 14 March 2026).

International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) (2025) *ICH E6(R3) Guideline for Good Clinical Practice (Step 4 Final Guideline)*. Available at: https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106_ErrorCorrections_2025_1024.pdf (Accessed: 18 February 2026).

IQVIA (2025) *IQVIA Laboratories launches Site Lab Navigator with breakthrough e-Requisition innovation to transform clinical trials* [Press release]. 4 March. Available at: https://www.iqvia.com/newsroom/2025/03/iqvia-laboratories-launches-site-lab-navigator-with-breakthrough (Accessed: 18 February 2026).

Kraaijenbrink, J. (2022) 'What is Industry 5.0 and how it will radically change your business strategy?', *Forbes*, 24 May. Available at: https://www.forbes.com/sites/jeroenkraaijenbrink/2022/05/24/what-is-industry-50-and-how-it-will-radically-change-your-business-strategy/ (Accessed: 18 February 2026).

Leslie, D. (2019) *Understanding artificial intelligence ethics and safety: A guide for the responsible design and implementation of AI systems in the public sector*. London: The Alan Turing Institute. Available at: https://doi.org/10.5281/zenodo.3240529

MarketsandMarkets (2025) *Central Lab Services Market: Global Forecast to 2030*. Available at: https://www.marketsandmarkets.com/Market-Reports/central-lab-services-market-106848348.html (Accessed: 18 February 2026).

McKinsey & Company (2025) *Generative AI in healthcare: Current trends and future outlook*. 26 March. Available at: https://www.mckinsey.com/industries/healthcare/our-insights/generative-ai-in-healthcare-current-trends-and-future-outlook (Accessed: 18 February 2026).

Mihic, A., Adabala Viswa, C., Agrawal, G., Yew, H. and Webster, K. (2025) *Unlocking peak operational performance in clinical development with artificial intelligence*. 9 January. Available at: https://www.mckinsey.com/industries/life-sciences/our-insights/unlocking-peak-operational-performance-in-clinical-development-with-artificial-intelligence (Accessed: 18 February 2026).

National Institute of Standards and Technology (NIST) (2023) *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. Gaithersburg, MD: National Institute of Standards and Technology. Available at: https://doi.org/10.6028/NIST.AI.100-1

Peres, R.S., Jia, X., Lee, J., Sun, K., Colombo, A.W. and Barata, J. (2020) 'Industrial artificial intelligence in Industry 4.0 – systematic review, challenges and outlook',

IEEE Access, 8, pp. 220121–220139. Available at: https://doi.org/10.1109/ACCESS.2020.3042874

Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (2024) *Official Journal L*, 2024/1689. Available at: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=OJ:L_202401689 (Accessed: 18 February 2026).

Schroer, C., Kruse, F. and Gomez, J.M. (2021) 'A systematic literature review on applying CRISP-DM process model', *Procedia Computer Science*, 181, pp. 526–534. Available at: https://doi.org/10.1016/j.procs.2021.01.199

Slope Clinical (2023) *The vicious cycle of clinical inventory management*. 4 April. Available at: https://www.slopeclinical.com/blog/the-vicious-cycle-of-clinical-inventory-management (Accessed: 18 February 2026).

Studer, S. *et al.* (2021) 'Towards CRISP-ML(Q): A machine learning process model with quality assurance methodology', *Machine Learning and Knowledge Extraction*, 3(2), pp. 392–413. Available at: https://doi.org/10.3390/make3020020

U.S. Food and Drug Administration (2023) *21 CFR Part 11: Electronic Records; Electronic Signatures*. Available at: https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-11 (Accessed: 18 February 2026).

AI Acknowledgement: Generative AI tools (Grammarly and Copilot) were used for proofreading and clarity refinement. The intellectual content remains entirely my own work.